

Investigating Visualization and Control for Assisting Gesture Interaction in Virtual Reality

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1 INTRODUCTION

Hand gesture interaction in Mixed Reality (MR) has become increasingly accessible thanks to improved hand tracking and software toolkits. This progress is driven by a clear motivation to extend hand interaction in MR, because our hands are natural and intuitive tools that we use in our everyday lives [3].

Current gesture interaction systems all rely on some form of gesture recognition to predict user's intents, and due to the limitation of current technology, recognition errors can occur. There are two types of recognition error: (1) the system falsely predicts a gesture when the user does not intend to do so, known as a false positive error; and (2) the system fails to recognize the gesture when the user attempts to perform a gesture, known as a false negative error. As discussed by Katsuragawa et al. [2], common methods of reducing false positive errors often involve tightening decision boundaries for positive samples, which results in more false negatives. This trade-off makes algorithmic improvements on both error types more difficult, resulting in imperfect prediction. Therefore, implementing some form of assistance is critical to reduce the impact of errors. However, there is limited work to date on studying what assistance strategies are most suitable in the context of hand gesture interaction.

The contributions of this work are two-fold: (1) we propose two pre-emptive techniques to provide assistance based on system uncertainty for gesture interaction in VR; and (2) we evaluate these techniques against a conventional menu and gained insights from the user performance and their feedback on controllability and visualization for assistance strategies.

2 METHODOLOGY

To investigate the preferred assistance strategy for VR gesture interaction, we adapted the system from Song et al. [5] to conduct the experiments.

2.1 Assistance Strategies

We designed three different strategies to provide assistance during gesture interaction, and they differ in terms of the level of control they give back to the user, and the amount of contextual information they contain about the system's decision.

1. **FIXED MENU:** A menu that is stationary and can be toggled on and off manually, and can be used to access all tools upon request by the user. The user can adjust the position of the menu by grabbing and dragging it.
2. **CONTEXTUAL MENU:** A pre-emptive menu which only appears when the level of uncertainty of the system's prediction

exceeds a certain threshold. After appearance the user can adjust its position and access the tools from the menu.

3. **VISUALIZATION:** A visual guide that appears based on the level of uncertainty in the same way as the **CONTEXTUAL MENU**. The guide includes a dynamic 3D demonstration of all gestures. The gesture with the highest likelihood based on the system prediction will be superimposed onto the user's hand location, while the rest of the gestures are displayed on the right side of the user's view, color-coded and ordered by probability.

2.2 Participants

We recruited 18 participants (average age = 26.7 ± 2.8 , min = 23, max = 32; 9 male, 9 female) for this study via advertising in a local university. The participants were asked to rate their prior VR experience. Three participants had never used VR before. Nine participants used VR for once or twice. Three participants used VR once a year. One participant used VR every three months, and two participants used VR once every month or more.

2.3 Procedure

At the start of the experiment, participants were shown a demonstration video of all nine gestures and were asked to memorize them. Then they were shown videos of the three assistance strategies with oral explanation. After the participants felt confident with the gestures and the assistance strategies, they were asked to complete a sequence of tasks, each with an instruction to use one specified 3D modelling tool. They could not proceed to the next task until they successfully finish the current task with the specified tool, which means if an error occurs they needed to attempt the task again. The instructions were randomized for each participant, and there were in total eight repetitions for each tool. The participants were told they could freely choose between menus and gestures during the menu conditions, as long as they complete the task as quickly and accurately as possible.

3 RESULTS

3.1 Performance

The average completion time for the **FIXED MENU**, **CONTEXTUAL MENU** and **VISUALIZATION** condition is 4.15, 3.98 and 4.36 seconds respectively. The average error rate for the three conditions are 6.48%, 5.02% and 4.32% respectively. Repeated-measure analysis of variance (RM-ANOVA) did not find statistical main effect between the different assistance strategies on the completion time. The non-parametric Friedman test did not find a main effect between the three assistance strategies on the selection error rates.

We separated the experiment into two blocks, each containing 12 individual tasks (3 gestures $\times$ 4 tasks each) to show the skill learning during the experiment. **Fig. 1a** displays the distribution of average completion time over the two blocks during the evaluation. **Fig. 1b** displays the distribution of average selection error rate across the two blocks of evaluation. The same statistical tests on each block did not show statistical difference between the three conditions, either.

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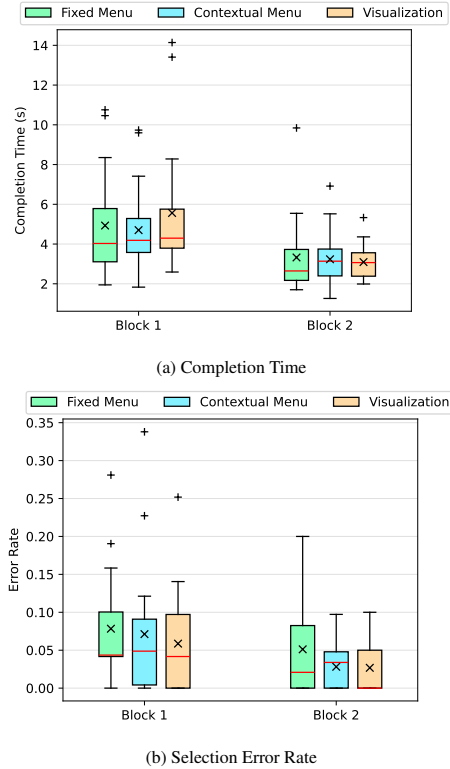


Figure 1: Distribution of a) completion time and b) selection error rate of the assistance strategies across the two blocks of evaluation. The boxplots show the median (the horizontal line), the mean ('x'), the first and third quartile (the box) and the minimum and maximum (the whiskers). The plus signs ('+') indicate outliers.

3.2 Preferences & Subjective Feedback

We asked participants to rank the three methods based on their overall preferences. Six out of the 18 participants preferred the FIXED MENU the most. Four participants preferred CONTEXTUAL MENU the most. Eight participants preferred VISUALIZATION the most. The distributions suggest that FIXED MENU and VISUALIZATION received comparable responses for user preference, while CONTEXTUAL MENU became the overall least preferred strategy.

Based on the performance and participant feedback, we summarize our key findings below.

3.2.1 Fixed Menu is Effective and Low-Risk

Given the experimental results, we consider the fixed menu to be the most preferred strategy. Not only did the fixed menu have marginal advantages in the agency, usability and workload ratings over the other two strategies, but it was also ranked in the first or second place by 16 participants. We attribute this broad acceptance to its high controllability and certainty, as the menu can be used at any-time and any location. We suggest that for an interaction task where the interaction goal is clearly defined, users prefer the method that assures the goal is achieved efficiently, regardless of the path that is taken to achieve the goal.

3.2.2 Visual Guidance Encourages Gesture Interaction

Unlike the menus, the VISUALIZATION method guides the user with visual information but forces them to entirely rely on the system's prediction. However, 14 participants ranked it in the first or

second place despite of its lack of control and relatively high frequency of occurrence. From the participant feedback we found the reasons to be two-fold: First, participants found the visual overlay to be easy to understand and follow, while the menus required extra effort to use. Secondly, participants demonstrated an interest in understanding the system's behavior.

Nonetheless, while some participants were fond of the visualizations, others did mention that the visual overlay was distracting, which shows the nuanced difference in what information format seems acceptable to individual users. We suggest that hand overlays should be applied with caution, and putting the visualization to the side can reduce distraction.

3.2.3 Contextual Information is Less Important than Controllability

In the guidelines from Amershi et al. [1] they pointed out that when a system is in doubt, it should "scope its service" and "make clear why", which correspond to the controllability and contextual information in this experiment. The participants' feedback for the CONTEXTUAL MENU clearly demonstrates the relative importance of giving control back to the user. The menu appearing unexpectedly incurred a cost that outweighed the assistance it was meant provide. The majority of participants preferred to decide when to seek help from a fixed menu, rather than receiving prompts from the system. This again shows that users value stability and controllability when error occurs, which agrees with the findings with Roy et al. [4].

4 CONCLUSION

In this work we have investigated three approaches to provide users with assistance during gesture interaction. The results and analysis show that when encountering difficulties, users prefer direct access to system functions upon their request rather than pre-emptive interfaces. It also shows that contextual information about the system's decision adds limited value to interaction, especially when the user already has direct access to system functions through a menu. On the other hand, users are more likely to attempt the gestures again when an error occurs, and visual guidance is useful to keep users engaged due to its simplicity and its ability to enhance user understanding about the system. We suggest to use direct control methods, such as standard menus for assisting gesture interaction, as they are well received by participants with no obvious disadvantages. A designer can consider adding visual guidance if users are motivated to adapt to the system and improve their proficiency.

REFERENCES

- [1] S. Amershi, D. Weld, M. Vorvoreanu, A. Fourney, B. Nushi, P. Collisson, J. Suh, S. Iqbal, P. N. Bennett, K. Inkpen, J. Teevan, R. Kikin-Gil, and E. Horvitz. Guidelines for human-ai interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, p. 1–13. Association for Computing Machinery, New York, NY, USA, 2019. doi: 10.1145/3290605.3300233 2
- [2] K. Katsuragawa, A. Kamal, Q. F. Liu, M. Negulescu, and E. Lank. Bi-level thresholding: Analyzing the effect of repeated errors in gesture input. *ACM Trans. Interact. Intell. Syst.*, 9(2–3), apr 2019. doi: 10.1145/3181672 1
- [3] D. A. Rosenbaum. *Human Motor Control*. Academic Press, 2009. 1
- [4] Q. Roy, F. Zhang, and D. Vogel. Automation accuracy is good, but high controllability may be better. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, p. 1–8. Association for Computing Machinery, New York, NY, USA, 2019. doi: 10.1145/3290605.3300750 2
- [5] Z. Song, J. J. Dudley, and P. O. Kristensson. Hotgestures: Complementing command selection and use with delimiter-free gesture-based shortcuts in virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 29(11):4600–4610, 2023. doi: 10.1109/TVCG.2023.3320257 1